```
import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import io
import os
from openai import OpenAI

# Page config
st.set_page_config(page_title="Data Analysis Agent", layout="wide")
st.title("📊 Data Analysis Agent")

# API Key setup
api_key = st.secrets.get("OPENAI_API_KEY") or os.getenv("OPENAI_API_KEY")

if api_key:
    client = OpenAI(api_key=api_key)
else:
    client = None
    st.warning("OpenAI API key not found. Q&A disabled.")

# File upload
uploaded_file = st.file_uploader("Upload CSV/TSV file", type=["csv", "tsv"])

if uploaded_file:
    sep = "\t" if uploaded_file.name.endswith("tsv") else ","
    df = pd.read_csv(uploaded_file, sep=sep)

    if len(df) > 10000:
        st.warning("Large dataset, sampling 10k rows")
        df = df.sample(10000, random_state=42)
```

```
st.success(f"Loaded: {df.shape[0]} rows, {df.shape[1]} columns")

option = st.sidebar.selectbox("Options", [
    "Preview Data", "Structure", "Summary",
    "Missing Values", "Visualizations", "Ask Questions"
])

if option == "Preview Data":
    st.dataframe(df.head(100))

elif option == "Structure":
    buffer = io.StringIO()
    df.info(buf=buffer)
    st.text(buffer.getvalue())

elif option == "Summary":
    st.dataframe(df.describe(include='all'))

elif option == "Missing Values":
    missing = df.isnull().sum()
    st.dataframe(missing)
    st.bar_chart(missing)

elif option == "Visualizations":
    num_cols = df.select_dtypes(include='number').columns

    if len(num_cols) == 0:
        st.warning("No numeric columns")
    else:
        chart = st.selectbox("Chart Type", ["Histogram", "Boxplot", "Heatmap"])
```

```
        if chart == "Histogram":
            col = st.selectbox("Column", num_cols)
            fig, ax = plt.subplots()
            ax.hist(df[col])
            st.pyplot(fig)

        elif chart == "Boxplot":
            col = st.selectbox("Column", num_cols)
            fig, ax = plt.subplots()
            sns.boxplot(x=df[col], ax=ax)
            st.pyplot(fig)

        elif chart == "Heatmap":
            fig, ax = plt.subplots()
            sns.heatmap(df[num_cols].corr(), annot=True, ax=ax)
            st.pyplot(fig)

elif option == "Ask Questions":
    question = st.text_input("Ask a question")

    if st.button("Ask"):
        if not client:
            st.error("API key missing")
        elif question:
            sample = df.head(50).to_string()

            response = client.chat.completions.create(
                model="gpt-5-mini",
                messages=[{
                    "role": "user",
```

```
            "content": f"Dataset:\n{sample}\n\nQuestion:\n{question}"
        }]
    )

    st.write(response.choices[0].message.content)
else:
    st.info("Upload a file to begin")
```